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COURSE: CHE 312: Transport Phenomena II

Fall Semester 2025

Department of Chemical Engineering

University of Illinois Chicago

Exam #3

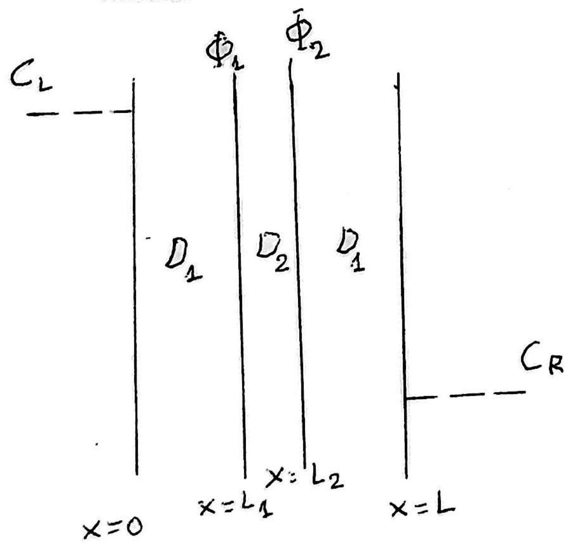
Duration: 70 minutes --- November 25, 2025

All Problems must be done

38

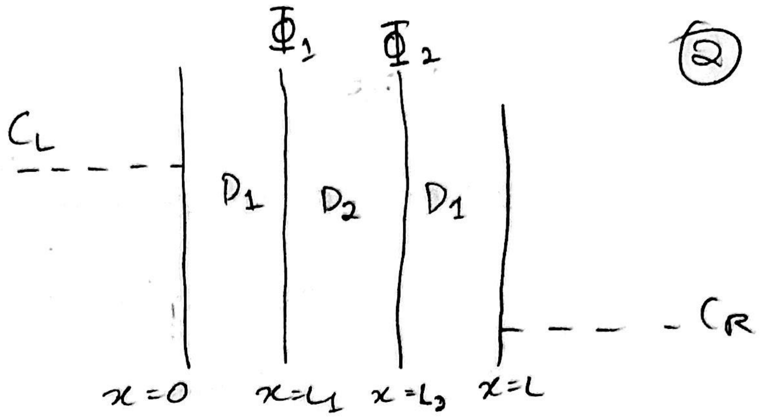
Problem 1: (30 points)
Consider steady state diffusion of the species A through three media arranged as shown in the attached diagram. Assume that diffusion is one-dimensional. At $x = 0$, $C_{A1} = C_L$, and at $x = L$, $C_{A3} = C_R$. In medium 2, there is a zeroth order chemical reaction through which the species A is consumed. Develop an expression for the concentration profiles across the three media.

(Bonus 10 points) Develop an expression for the steady state flux across the three media.



Given:

- Steady state diffusion
- 1-D
- At $x=0$, $C_{A1} = C_{L1}$
- At $x=L$, $C_{A3} = C_R$
- Medium 2 → zeroth order



- Medium 1 $(0 \le x \le L_1) \rightarrow D_1$
- Medium 2 $(L_1 \le x < L_2) \rightarrow D_2$
- Medium 3 $(L_2 < x\ L) \rightarrow D_3$

B.C: · At $x=0$, $C_{A1} = C_L$

· At $x=L$, $C_{A3} = C_R$

At steady state the flux is the same everywhere

$N_A = \Phi$

Medium 1 (no reaction) $= D_1 \dfrac{d^2 C_1}{dx^2} = 0 \;=\; C_1(x) = A_1 x + B_1$

Medium 2 (zero-order) $= D_2 \dfrac{d^2 C_2}{dx^2} + K = 0 \Rightarrow \dfrac{d^2 C_2}{dx^2} = \dfrac{-K}{D_2}$

$\Rightarrow \dfrac{dC_2}{dx} = \dfrac{-K}{D_2} x + A_2$

$\Rightarrow C_2(x) = \dfrac{-K}{2D_2} x^2 + A_2 x + B_2$

BC1: at $x=0$, $C_{A1} = C_{L1} \Rightarrow A_1(C_{L1}) + B_1 C_{L1}$

BC2: at $x=L$; $C_{A3} = C_R \Rightarrow \dfrac{-K}{2D_2} L^2 + A_2 L + B_2 = C_R$

continuity of concentration at $x=0$: $C_1(0) = C_2(0) \Rightarrow B_1 = B_2 = C_0$

continuity of flux at $x=0$: $D_1 \dfrac{dC_1}{dx}\Big|_0 = -D_2 \dfrac{dC_2}{dx}\Big|_0$

$\Rightarrow -D_1 A_1 = -D_2 A_2 \Rightarrow D_1 A_1 = D_2 A_2$

From boundary condition, $A_2 = \dfrac{D_1}{D_2} A_1$

BC1: $B_1 = C_0 \Rightarrow C_0 = C_L + A_1 L$

BC2: $B_2 = C_0 \Rightarrow C_R = \dfrac{K}{2D_2} L^2 + A_2 L + C_0$

$$\Rightarrow C_R = \frac{-k}{2D_2} L^2 + \frac{D_1}{D_2}$$

$$\Rightarrow A_1 L + C_L + A_1 L \Rightarrow A_1 (L)\left(1 + \frac{D_1}{D_2}\right) = C_R - C_L + \frac{kL^2}{2D_2}$$

$$* \Rightarrow A_1 = \frac{D_2}{L(D_1 + D_2)} \left(C_R - C_L + \frac{kL^2}{2D_2} \right) = \frac{D_2}{L} \frac{(C_R - C_L)}{L(D_1 + D_2)} + \frac{kL}{2(D_1 - D_2)}$$

$$* \Rightarrow A_2 = \frac{D_1}{D_2} A_1 \,, \quad C_0 = C_L + A_1 L = \frac{C_L + D_2 (C_R - C_L)}{D_1 + D_2} + \frac{kL^2}{2(D_1 + D_2)}$$

Medium 1: $0 \leq x \leq L_1 \Rightarrow C_1(x) = A_1(x) + C_0$

Medium 2: $L_1 \leq x \leq L_2 \Rightarrow C_2(x) = \frac{-k}{2D_2} x^2 + A_2 x + C_0$

Medium 3: $L_2 \leq x \leq L \Rightarrow C_3(x) = A_3(x) + C_0$

Steady State flux $\Rightarrow$ Fick's Law

Medium 1 (no reaction): $N_A = \phi = -D_1 \frac{dc_1}{dx} = -D_1 A_1$

$$\Phi = -D_1 \left[\frac{D_2}{L} \frac{(C_R - C_L)}{(D_1 + D_2)} + \frac{kL}{2(D_1 + D_2)} \right] = -\frac{D_1 D_2}{L(D_1 + D_2)} (C_R - C_L) - \frac{D_1 kL}{2(D_1 + D_2)}$$

(4) CONT. FROM PROBLEM 2, PART 2

$\Rightarrow C_A(y) = A\cosh(\lambda y) - \dfrac{k_1}{k_{-1}}$

B.C: at $y = L$

$C_A(L) = C_L$, $C_L = A\cosh(\lambda L) - \dfrac{k_1}{k_{-1}}$

$\Rightarrow A = \dfrac{C_L + \frac{k_1}{k_{-1}}}{\cosh(\lambda L)}$, $\lambda = \sqrt{k_{-1}/D}$

$\Rightarrow C_A(y) = \left(C_L + \dfrac{k_1}{k_{-1}}\right) \dfrac{\cosh\left(\sqrt{\frac{k_{-1}}{D}}\, y\right)}{\cosh\left(\sqrt{\frac{k_{-1}}{D}}\, L\right)} - \dfrac{k_1}{k_{-1}}$

Problem 2: (70 points)

Consider the reversible conversion of species A into B through the stoichiometric reactions $A \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} B$ in a stationary liquid film of thickness L. Determine C_A and C_B throughout the film, if

35 (35 Points) the surface at y = L is assumed to be impermeable and unreactive, and the concentrations of A and B at y = 0 are specified as C_{A0} and C_{B0}. The forward and reverse homogeneous reactions follow zeroth order kinetics

35 (35 Points) the surface at y = L is assumed to be impermeable and unreactive, and the concentrations of A and B at y = 0 are specified as C_{A0} and C_{B0}. The forward and reverse homogeneous reactions follow zeroth and first order kinetics, respectively

1) Given:
- 1D
- Steady diffusion of A in a stationary liquid film of thickness L, $0 \le y \le L$
- $R_{AB} = K_1$, $R_{BA} = -K_1$ (both zeroth order kinetics)

Rate (A) = $r_A = K_1 + k_{-1}$

General steady state diffusion reaction equation for A in 1D: $D \dfrac{d^2 C_A}{dy^2} - r_A = 0$

B.C: impermeable, unreactive wall at $y = 0$

$$\Rightarrow N_A = \frac{-D\,dC_A}{dy} = 0 \Rightarrow \left.\frac{dC_A}{dy}\right|_{y=0} = 0$$

specified concentration at outer surface $y = L$

$C_A(L) = C_L$, $r_A = k_1 + K_{-1} = K_T$ (constant)

$$D\frac{d^2 C_A}{dy^2} - K_T = 0 \Rightarrow \frac{d^2 C_A}{dy^2} = \frac{K_T}{D} = K$$

$$\Rightarrow \frac{dC_A}{dy} = ky + C_1$$

Applying BC1 at $y=0$,

$$\frac{dC_A}{dy}=0 \Rightarrow C_1=0$$

$$\frac{dC_A}{dy}=Ky \Rightarrow C_A(y)=\frac{K}{2}y^2+C_2$$

BC at $y=L$,

$$C_A(L)=C_L \Rightarrow C_L=\frac{K}{2}y^2+C_2 \Rightarrow C_2=C_L-\frac{K}{2}L^2$$

$$K=\frac{k_1+k_{-1}}{D}$$

$$\Rightarrow C_A(y)=C_L-\frac{k_1-k_{-1}}{2D}(c^2-y^2)$$

2) $R_{AB}=k_1$ (zeroth order), $R_{BA}=-k_{-1}C_A$ (first order)

$$\Rightarrow r_A=k_1+k_{-1}C_A \Rightarrow D\frac{d^2C_A}{dy^2}-(k_1+k_{-1}C_A)=0$$

$$\Rightarrow \frac{d^2C_A}{dy^2}-\frac{k_{-1}}{D}C_A=\frac{k_1}{D}, \quad \text{Let } \lambda^2=\frac{k_{-1}}{D}$$

$$\Rightarrow \frac{d^2C_A}{dy^2}-\lambda^2C_A=\frac{k_1}{D}$$

Homogeneous : $C_h(y)=A\cosh(\lambda y)+B\sinh(\lambda y)$

$C_p=C \Rightarrow 0-\lambda^2C^*=\frac{k_1}{D} \Rightarrow C^*=\frac{-k_1}{-k_{-1}}$

$$\Rightarrow C_A(y)=A\cosh(\lambda y)+B\sinh(\lambda y)-\frac{k_1}{k_{-1}}$$

B.C $y=0$, $\frac{dC_A}{dy}=A\lambda\sin(\lambda y)+B\lambda\cosh(\lambda y)$

$y=0$: $\sinh 0=0$, $\cosh 0=1$

$$\Rightarrow \left.\frac{dC_A}{dy}\right|_0=B\lambda=0 \Rightarrow B=0$$

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(4) CONT. FROM PROBLEM 2, PART 2

$$\Rightarrow C_A(y) = A\cosh(\lambda y) - \frac{k_1}{k_{-1}}$$

B.C: at $y = L$

$$C_A(L) = C_L, \quad C_L = A\cosh(\lambda L) - \frac{k_1}{k_{-1}}$$

$$\Rightarrow A = \frac{C_L + \frac{k_1}{k_{-1}}}{\cosh(\lambda L)}, \quad \lambda = \sqrt{k_{-1}/D}$$

$$\Rightarrow C_A(y) = \left(C_L + \frac{k_1}{k_{-1}}\right) \frac{\cosh\left(\sqrt{\frac{k_{-1}}{D}}\, y\right)}{\cosh\left(\sqrt{\frac{k_{-1}}{D}}\, L\right)} - \frac{k_1}{k_{-1}}$$